MINOR PROJECT - II

BRAIN TUMUOR DETECTION SYSTEM DOCUMENTATION

Submitted By: -

Livingstone Mazvovere - 20222730

B.voc Software Development

Exam Roll: 22020107055

GitHub Link: click here !

Live Project: click here !

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1. Introduction.

This project is an AI-powered Brain Tumour Detection System that leverages deep learning techniques to analyse MRI scans and detect potential brain tumors. The system provides an easy-to-use interface where users can upload MRI images, receive instant analysis, and generate a downloadable PDF report with the results.

2. Problem Statement.

Brain tumors are life-threatening conditions that require **earl**y and **accurat**e diagnosis. Manual inspection of MRI scans is time-consuming and may be prone to human errors. This system aims to automate the detection process, reducing diagnostic time and improving accuracy.

3. Objectives.

1. Automate the detection of brain tumors using deep learning.
2. Develop a user-friendly interface for clinicians.
3. Provide reliable, real-time detection with performance metrics.
4. Generate professional diagnostic reports.

4. System Overview.

The system allows users to upload **MRI scans**, processes the images using a fine-tuned **YOLOv11m** model, highlights tumors, and generates diagnostic reports. Built with **Streamlit**, **PyTorch**, **OpenCV**, and **FPDF**.

5. Project Timeline.

Phase 1: Requirement Analysis and Data Collection	Week 1
Phase 2: Data Preparation - Cleaning, Preprocessing, and Augmentation	Week 2
Phase 3: Model Development - Fine-tuning YOLOv11 for tumor detection and testing	Week 3
Phase 4: System Design - Implementing Streamlit UI and integrating FPDF for report generation	Week 4
Phase 5: Deployment - Deploying on Streamlit Cloud and performing functional testing	Week 5
Phase 6: Final Testing and Delivery - Validating system performance, refining UI, and generating reports	Week 6

6. Technologies Used.

1. **Streamlit:**
   - For building the web interface.
2. **OpenCV:**
   - For image processing and pre-processing.
3. **NumPy:**
   - For handling numerical operations.
4. **YOLO (Ultralytics):**
   - Deep learning model for tumour detection.
5. **Pillow (PIL):**
   - For handling image formats.
6. **FPDF:**
   - For generating PDF reports.

7. Installation and Setup.

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1. **Clone the repository.**
   - git clone https://github.com/PresidentLivingstone/Brain_tumor_detection-system-using-fined-tuned-yolov11.git

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2. **Create a virtual environment (MacOS).**

   - python3 -m venv venv
   - source venv/bin/activate

---

3. **Install dependencies.**

   - pip install -r requirements.txt

---

4. **Run.**
   - Streamlit run app.py

---

8. Model Training

Brain Tumor Detection and Localization Using YOLO v1[redacted] and Deep Learning Techniques in Python

```
# check for GPU availability
!nvidia-smi
```

```
Tue Feb 25 11:22:36 2025
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 550.54.15              Driver Version: 550.54.15      CUDA Version: 12.4     |
|-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  Tesla T4                       Off |   00000000:00:04.0 Off |                    0 |
| N/A   70C    P8             11W /   70W |       0MiB /  15360MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI        PID   Type   Process name                              GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|  No running processes found                                                             |
+-----------------------------------------------------------------------------------------+
```

```
import torch
torch.cuda.empty_cache()
```

Step 01 # Install the Ultralytics Package

```
# install ultralytics
!pip install ultralytics
```

Show hidden output

Step 02 # Import All the Requried Libraries

```
# import libraries
import ultralytics
ultralytics.checks()
from ultralytics import YOLO
from IPython.display import Image
```

```
Ultralytics 8.3.78 🚀 Python-3.11.11 torch-2.5.1+cu124 CUDA:0 (Tesla T4, 15095MiB)
Setup complete ✅ (2 CPUs, 12.7 GB RAM, 33.2/112.6 GB disk)
```

Step # 03 Download Dataset from Roboflow

```
# load the dataset
!pip install roboflow

from roboflow import Roboflow
rf = Roboflow(api_key="0zA9553LIS7TIrJO3Cr4")
project = rf.workspace("aykut-izfoo").project("brain-tumor-detection-rwfby")
version = project.version(3)
dataset = version.download("yolov11")
```

Step # 04 Train YOLOv12 Model on a Custom Dataset

```
!yolo task=detect mode=train data={dataset.location}/data.yaml model="yolo11m.pt" epochs=20 imgsz=640 batch=8
```

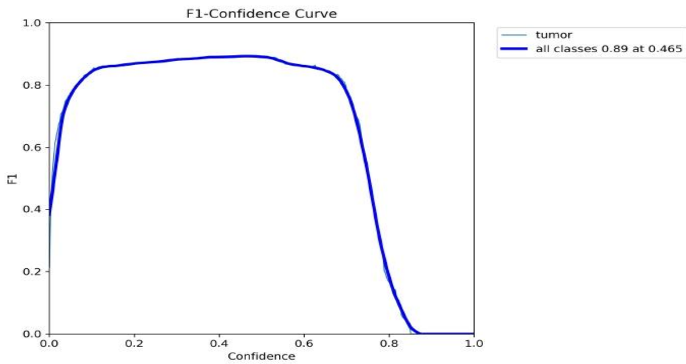
```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      12/20      7.79G      1.387     0.9343      1.569          8        640: 100% 150/150 [01:53<00:00,
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% 10/10 [00:0
                   all        150        159      0.854      0.874       0.89      0.484

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      13/20      7.79G      1.367     0.9299      1.575          8        640: 100% 150/150 [01:53<00:00,
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% 10/10 [00:0
                   all        150        159      0.873       0.83      0.877      0.507

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      14/20      7.79G      1.337     0.8876      1.517         10        640: 100% 150/150 [01:53<00:00,
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% 10/10 [00:0
                   all        150        159      0.881      0.843      0.871      0.516

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      15/20      7.74G      1.296     0.8434      1.486         13        640: 100% 150/150 [01:53<00:00,
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% 10/10 [00:0
                   all        150        159      0.863      0.843      0.892       0.53

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      16/20      7.79G      1.287     0.8211      1.487          8        640: 100% 150/150 [01:53<00:00,
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% 10/10 [00:0
                   all        150        159      0.925      0.824       0.88      0.532

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      17/20      7.79G      1.267     0.7794      1.473          8        640: 100% 150/150 [01:53<00:00,
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% 10/10 [00:0
                   all        150        159      0.911      0.874      0.921      0.552

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      18/20      7.79G      1.248     0.7439       1.46          9        640: 100% 150/150 [01:53<00:00,
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% 10/10 [00:0
                   all        150        159      0.892      0.883      0.895      0.538

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      19/20      7.74G      1.244       0.73       1.44          8        640: 100% 150/150 [01:53<00:00,
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% 10/10 [00:0
                   all        150        159      0.876       0.89      0.904       0.55

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      20/20      7.79G      1.202     0.7193      1.413          8        640: 100% 150/150 [01:53<00:00,
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% 10/10 [00:0
                   all        150        159      0.908      0.874       0.89      0.549

20 epochs completed in 0.666 hours.
Optimizer stripped from runs/detect/train/weights/last.pt, 40.5MB
Optimizer stripped from runs/detect/train/weights/best.pt, 40.5MB

Validating runs/detect/train/weights/best.pt...
Ultralytics 8.3.78 🚀 Python-3.11.11 torch-2.5.1+cu124 CUDA:0 (Tesla T4, 15095MiB)
YOLO11m summary (fused): 125 layers, 20,030,803 parameters, 0 gradients, 67.6 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% 10/10 [00:0
                   all        150        159      0.914      0.874      0.921      0.552
Speed: 0.3ms preprocess, 24.9ms inference, 0.1ms loss, 2.4ms postprocess per image
Results saved to runs/detect/train
💡 Learn more at https://docs.ultralytics.com/modes/train
```

Analysis:

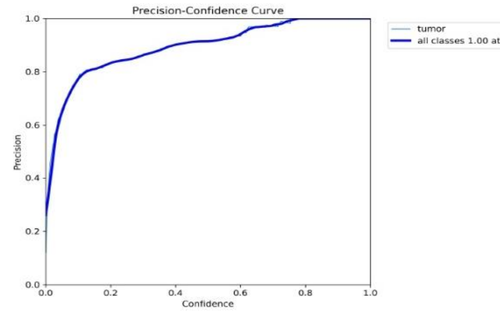
1. YOLO 11m.pt model trained for **20 epochs** on a brain tumor dataset in **0.668 hours.**
2. Losses steadily decreased: **box loss to 1.202, cls_loss to 0.7193, dfl_loss to 1.413.**
3. **Final mAP50 reached 100%, and mAP50-95 was 0.552**, showing strong detection performance.
4. **Precision and Recall were 0.914 and 0.874** respectively, indicating good model accuracy.
5. Best model saved to **runs/detect/train/weights/best.pt.**

Step # 05 Examine Training Results

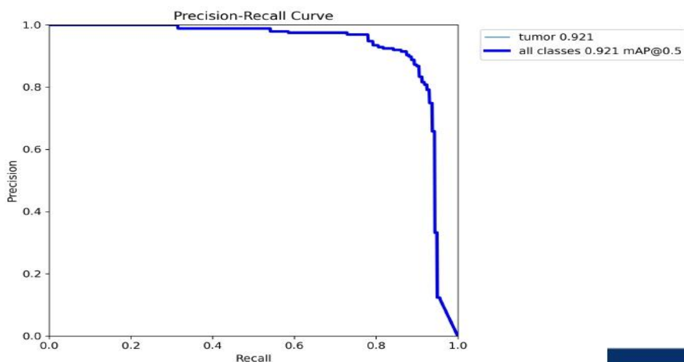
```
Image("/content/runs/detect/train/F1_curve.png", width=600)
```



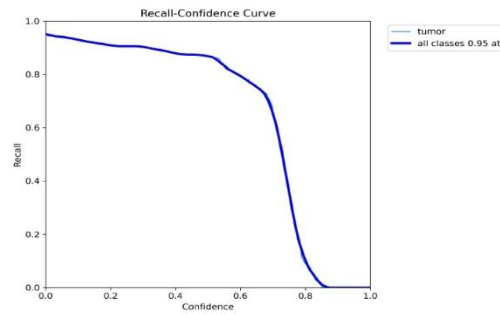

```
Image("/content/runs/detect/train/P_curve.png", width=600)
```



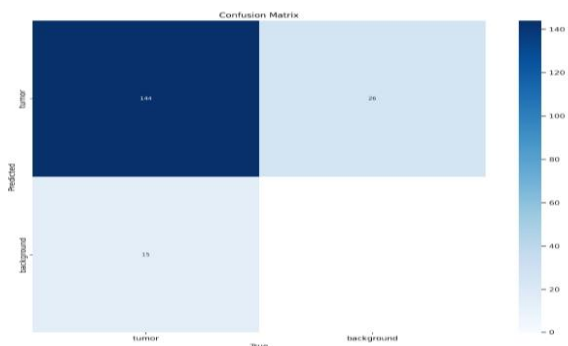

```
Image("/content/runs/detect/train/PR_curve.png", width=600)
```




```
Image("/content/runs/detect/train/R_curve.png", width=600)
```




```
Image("/content/runs/detect/train/confusion_matrix.png", width=600)
```



Analysis:

1. **F1-Confidence Curve**: The F1 score peaks **at 0.89 around 0.465 confidence**, indicating optimal balance between precision and recall at that point.
2. **Precision-Recall Curve**: Precision remains **high (0.921)** across most recall values, showing reliable detection of tumor instances.
3. **Precision-Confidence Curve**: Precision reaches a perfect score of **1.00 at high** confidence thresholds, confirming very few false positives.
4. **Recall-Confidence Curve**: Recall peaks **at 0.95** but drops sharply at higher confidence, indicating missed detections at strict thresholds.
5. **Confusion Matrix**: The model correctly **predicted 144 tumor instances.** It **made 41 incorrect predictions** (**15 false negatives and 26 false positives**). This indicates strong classification performance with room for slight improvement.

Step # 06 Validate Fine-Tuned Model

```
!yolo task=detect mode=val model="/content/runs/detect/train/weights/last.pt" data={dataset.location}/data.yam
```

```
Ultralytics 8.3.78 🚀 Python-3.11.11 torch-2.5.1+cu124 CUDA:0 (Tesla T4, 15095MiB)
YOLO11m summary (fused): 125 layers, 20,030,803 parameters, 0 gradients, 67.6 GFLOPs
val: Scanning /content/Brain-Tumor-Detection-3/valid/labels.cache... 150 images, 0 backgrounds, 0 corrupt:
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% 10/10 [00:0
                   all        150        159      0.908      0.874       0.89      0.549
Speed: 4.2ms preprocess, 23.7ms inference, 0.0ms loss, 2.0ms postprocess per image
Results saved to runs/detect/val
💡 Learn more at https://docs.ultralytics.com/modes/val
```

Analysis:

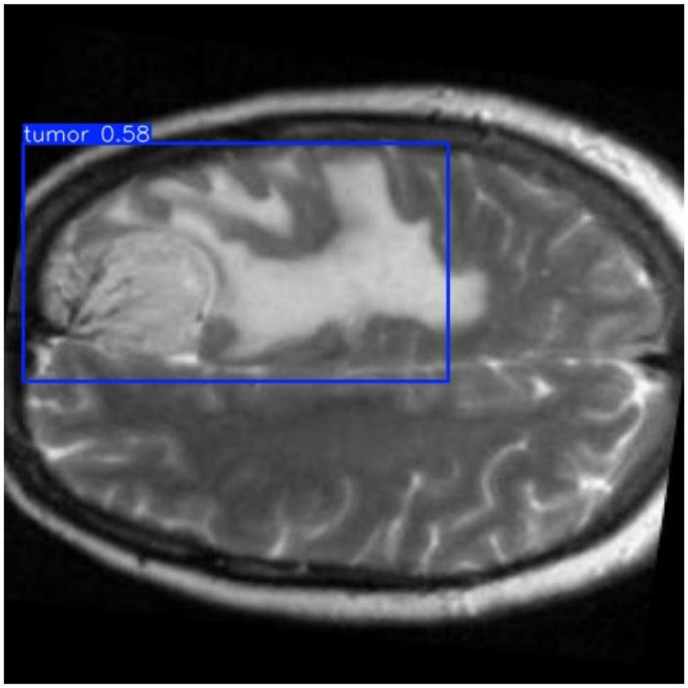
1. The fine-tuned YOLOv11 model was validated on 150 images with 159 tumor instances.
2. It achieved a **Precision (P)** of 0.908 and **Recall (R)** of 0.874.
3. The **mAP@0.5** was 0.89, indicating high detection accuracy.
4. The **mAP@0.5:0.95** score of 0.549 shows decent performance across stricter IoU thresholds.
5. Validation was fast, with 23.7ms inference time per image.

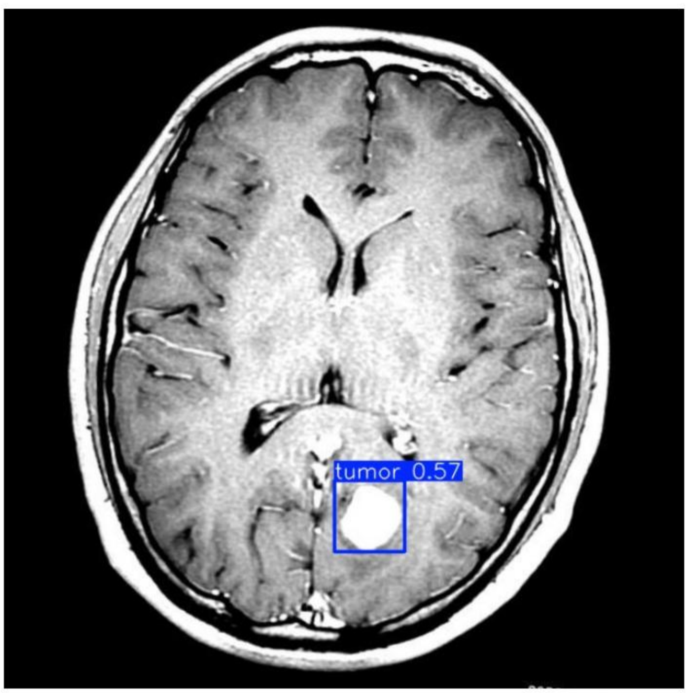
Step # 07 Inference with Custom Model on Images

```
!yolo task=detect mode=predict model="/content/runs/detect/train/weights/best.pt" conf=0.25 source={dataset.lo
```

```
image 96/150 /content/Brain-Tumor-Detection-3/test/images/i588_jpg.rf.6b3391fa86a9d4aff82a8f521391368c.jpg
image 97/150 /content/Brain-Tumor-Detection-3/test/images/i58_jpg.rf.8506a4568133f10cd507adef43982620.jpg:
image 98/150 /content/Brain-Tumor-Detection-3/test/images/i623_jpg.rf.9bdcbcabfcb5e51bd5a2724771e80d98.jpg
image 99/150 /content/Brain-Tumor-Detection-3/test/images/i632_jpg.rf.297489cc9d56e7d262e83545e977ea7a.jpg
image 100/150 /content/Brain-Tumor-Detection-3/test/images/i637_jpg.rf.46c2211bedfd6f6cedfd715855c640da.jp
image 101/150 /content/Brain-Tumor-Detection-3/test/images/i638_jpg.rf.76b6088d4ba38314b4372a6d2efbca3b.jp
image 102/150 /content/Brain-Tumor-Detection-3/test/images/i641_jpg.rf.a4954f7f47ec8613bae4a57cbce730e0.jp
image 103/150 /content/Brain-Tumor-Detection-3/test/images/i653_jpg.rf.e4e6062de8130ad6d2038151f592f23a.jp
image 104/150 /content/Brain-Tumor-Detection-3/test/images/i666_jpg.rf.23aa2b1551189b8ad68e77b03d12f545.jp
image 105/150 /content/Brain-Tumor-Detection-3/test/images/i669_jpg.rf.4e56b286ed4308928dbd96b50c25ef3e.jp
image 106/150 /content/Brain-Tumor-Detection-3/test/images/i675_jpg.rf.224e3e23da129a4a0b5b3a098dd16ec4.jp
image 107/150 /content/Brain-Tumor-Detection-3/test/images/i683_jpg.rf.b2dfef6ca9297f9364b618e982144f75.jp
image 108/150 /content/Brain-Tumor-Detection-3/test/images/i694_jpg.rf.13e27026ee0030b9053d430d68152c1a.jp
image 109/150 /content/Brain-Tumor-Detection-3/test/images/i712_jpg.rf.567d7079fdcbf99d48d17867278527d8.jp
image 110/150 /content/Brain-Tumor-Detection-3/test/images/i724_jpg.rf.4c51b2c0c5719bdb454f02d9829a58f8.jp
image 111/150 /content/Brain-Tumor-Detection-3/test/images/i726_jpg.rf.a5664c07c58928c8d2c9574ca752a33d.jp
image 112/150 /content/Brain-Tumor-Detection-3/test/images/i734_jpg.rf.099f749013745047d94661a66722a9a9.jp
image 113/150 /content/Brain-Tumor-Detection-3/test/images/i737_jpg.rf.aff5ce741a41fc19e86202e7e31f82c3.jp
image 114/150 /content/Brain-Tumor-Detection-3/test/images/i747_jpg.rf.5f606b9859adebf1af26ed621f4bb592.jp
image 115/150 /content/Brain-Tumor-Detection-3/test/images/i748_jpg.rf.b17f73f2c20648630ace8d65018b0aa1.jp
image 116/150 /content/Brain-Tumor-Detection-3/test/images/i758_jpg.rf.e88ebfc5c4dc3ca39f347a8fee35ac87.jp
image 117/150 /content/Brain-Tumor-Detection-3/test/images/i761_jpg.rf.02f776ba1f0d601e560792735cdc40b0.jp
image 118/150 /content/Brain-Tumor-Detection-3/test/images/i771_jpg.rf.4207b274e51e172c643ea7948cfea397.jp
image 119/150 /content/Brain-Tumor-Detection-3/test/images/i789_jpg.rf.ea9b8127a52fb24ea03907e4fbcf4147.jp
image 120/150 /content/Brain-Tumor-Detection-3/test/images/i805_jpg.rf.a955bbb1b73ab245524d89ced3a0563f.jp
image 121/150 /content/Brain-Tumor-Detection-3/test/images/i807_jpg.rf.314992ca981c655d2eea5b7cfe2254de.jp
image 122/150 /content/Brain-Tumor-Detection-3/test/images/i819_jpg.rf.cc3e33a1726054de9372342b2ae4eec8.jp
image 123/150 /content/Brain-Tumor-Detection-3/test/images/i827_jpg.rf.324fc10d039bbebb131b342def5b873a.jp
image 124/150 /content/Brain-Tumor-Detection-3/test/images/i828_jpg.rf.62524a984eaa937c24d2fc5c66ed3a19.jp
image 125/150 /content/Brain-Tumor-Detection-3/test/images/i82_jpg.rf.b3eb7b97647849aca8c8cfaac3bdd991.jpg
image 126/150 /content/Brain-Tumor-Detection-3/test/images/i833_jpg.rf.4e9845b2514a319c89faca07cf68b279.jp
image 127/150 /content/Brain-Tumor-Detection-3/test/images/i843_jpg.rf.a59ba973eeb237c90553538e26c12472.jp
image 128/150 /content/Brain-Tumor-Detection-3/test/images/i846_jpg.rf.d24a8bf1cc973173c469ea8e72a4c14f.jp
image 129/150 /content/Brain-Tumor-Detection-3/test/images/i857_jpg.rf.02ee23f4c9397045a1c3d960b79b24b5.jp
image 130/150 /content/Brain-Tumor-Detection-3/test/images/i868_jpg.rf.c8a0b2fe11eb41a75b18ad55d212fd89.jp
image 131/150 /content/Brain-Tumor-Detection-3/test/images/i884_jpg.rf.4fcd40fe90b3926fea7294c4ffb1e61c.jp
image 132/150 /content/Brain-Tumor-Detection-3/test/images/i896_jpg.rf.4c5b309df255e5688366999b08e8e34e.jp
image 133/150 /content/Brain-Tumor-Detection-3/test/images/i908_jpg.rf.12514bd53bedd5b736af703e79f3f1be.jp
image 134/150 /content/Brain-Tumor-Detection-3/test/images/i910_jpg.rf.33913149831b29dd515fdbc9446edb1c.jp
image 135/150 /content/Brain-Tumor-Detection-3/test/images/i912_jpg.rf.8f72c97302d1dbc224597f796c61a8ca.jp
image 136/150 /content/Brain-Tumor-Detection-3/test/images/i926_jpg.rf.861915433eedb4352757f3d0d6b2b6db.jp
image 137/150 /content/Brain-Tumor-Detection-3/test/images/i935_jpg.rf.95016f5f65086d3a8c3e2a62672bed1a.jp
image 138/150 /content/Brain-Tumor-Detection-3/test/images/i941_jpg.rf.08a84ea2cc0abaad980adb41ff4bf342.jp
image 139/150 /content/Brain-Tumor-Detection-3/test/images/i965_jpg.rf.52bd1c05abbc6fe082cfb8c5a70b3f36.jp
image 140/150 /content/Brain-Tumor-Detection-3/test/images/i969_jpg.rf.c441e283f4968ef217dfd515d88b3e3e.jp
image 141/150 /content/Brain-Tumor-Detection-3/test/images/i96_jpg.rf.7ca0d84fc0783bba267f125e4ed1c620.jpg
image 142/150 /content/Brain-Tumor-Detection-3/test/images/i970_jpg.rf.174ea2ccc37f0523214af069eb705615.jp
image 143/150 /content/Brain-Tumor-Detection-3/test/images/i983_jpg.rf.eb2a296d825cde6962d76a841d0f5d12.jp
image 144/150 /content/Brain-Tumor-Detection-3/test/images/i987_jpg.rf.3f5bf6959857deed8375852d6d977f2d.jp
image 145/150 /content/Brain-Tumor-Detection-3/test/images/i988_jpg.rf.a30953ab658ccbd589aec95987ca4650.jp
image 146/150 /content/Brain-Tumor-Detection-3/test/images/i98_jpg.rf.b0c828ac25c2196641290302201bcf6f.jpg
image 147/150 /content/Brain-Tumor-Detection-3/test/images/i990_jpg.rf.4bce0a3e80e3ef872db7d8c055718634.jp
image 148/150 /content/Brain-Tumor-Detection-3/test/images/i992_jpg.rf.cffa34774438e2aee449765a6a680757.jp
image 149/150 /content/Brain-Tumor-Detection-3/test/images/i993_jpg.rf.e5ba6e0bcae9fc7be2ee42f17981eb18.jp
image 150/150 /content/Brain-Tumor-Detection-3/test/images/i995_jpg.rf.2fdc64de1cf961e2344e4f9f42961e29.jp
Speed: 2.1ms preprocess, 20.3ms inference, 2.2ms postprocess per image at shape (1, 3, 640, 640)
Results saved to runs/detect/predict
💡 Learn more at https://docs.ultralytics.com/modes/predict
```

```
import glob
import os
from IPython.display import Image as IPyImage, display

latest_folder = max(glob.glob('/content/runs/detect/predict*/'), key=os.path.getmtime)
for img in glob.glob(f'{latest_folder}/*.jpg')[1:4]:
    display(IPyImage(filename=img, width=600))
    print("\n")
```


tumor 0.58


tumor 0.57

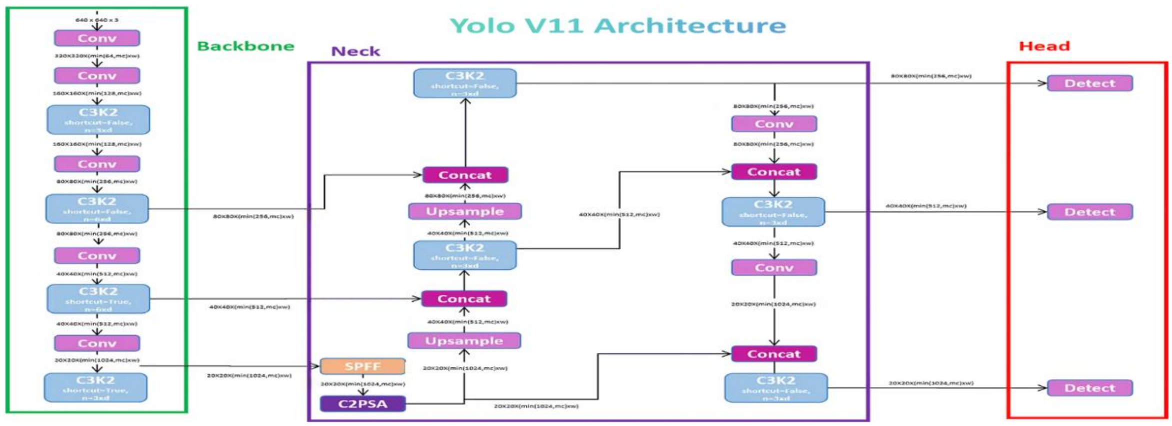
Step #08 Save the Model in torch Script format for easy compatibility

```
# convert  model to torchscript
model = YOLO("/content/runs/detect/train/weights/best.pt")
model.export(format="torchscript")
```

```
Ultralytics 8.3.78 🚀 Python-3.11.11 torch-2.5.1+cu124 CPU (Intel Xeon 2.00GHz)
YOLO11m summary (fused): 125 layers, 20,030,803 parameters, 0 gradients, 67.6 GFLOPs

PyTorch: starting from '/content/runs/detect/train/weights/best.pt' with input shape (1, 3, 640, 640) BCHW

TorchScript: starting export with torch 2.5.1+cu124...
TorchScript: export success ✅ 7.4s, saved as '/content/runs/detect/train/weights/best.torchscript' (77.1 I

Export complete (10.9s)
Results saved to /content/runs/detect/train/weights
Predict:         yolo predict task=detect model=/content/runs/detect/train/weights/best.torchscript imgsz=(
Validate:        yolo val task=detect model=/content/runs/detect/train/weights/best.torchscript imgsz=640 (
Visualize:       https://netron.app
'/content/runs/detect/train/weights/best.torchscript'
```

```
import cv2
import torch
import matplotlib.pyplot as plt
from ultralytics import YOLO

def predict_and_display_v11(image_path, model_path, confidence_threshold=0.5, target_size=(640, 640)):
    """
    Detects and visualizes brain tumor presence in an image using a fine-tuned YOLOv11 model.
    Enhances the image by resizing and converting it to grayscale before prediction.

    Args:
        image_path: Path to the input image.
        model_path: Path to the trained YOLOv11 TorchScript model.
        confidence_threshold: Minimum confidence score for a detection to be considered.
        target_size: The target size for resizing the image.
    """
    # Load and preprocess the image
    img = cv2.imread(image_path)
    img_resized = cv2.resize(img, target_size)
```

9. Yolov11 Architecture.



YOLOv11 Model Architecture

YOLOv11, the latest in the YOLO series, enhances real-time object detection with improved speed and accuracy……... Source

Architecture Highlights:

- **Backbone:** Employs Conv Block, Bottle Neck, and the efficient C3K2 block for feature extraction, replacing the C2F block of YOLOv8.
- **Neck:** Retains SPFF (Spatial Pyramid Pooling Fast) for multi-scale feature pooling, crucial for detecting objects of varying sizes.
- **Attention Mechanism:** Introduces the C2PSA block with spatial attention to focus on important image regions, improving detection of small or occluded objects.
- **Head:** Utilizes a multi-scale prediction head to detect objects at different sizes using feature maps from the backbone and neck.

Key Architectural Innovations:

- **C3K2 Block:** An evolution of CSP bottleneck, using smaller 3x3 convolutions for faster and more efficient computation while preserving feature capture.
- **SPFF:** Aggregates multi-scale contextual information, enhancing the detection of small objects.
- **C2PSA:** Incorporates position-sensitive attention to emphasize spatially relevant features.

Performance Metrics:

- **mAP (Mean Average Precision):** Measures the balance between precision and recall across classes and IoU thresholds.
- **IoU (Intersection Over Union):** Quantifies the overlap between predicted and ground truth bounding boxes.
- **FPS (Frames Per Second):** Indicates the model's processing speed, vital for real-time applications.

10. System Architecture.

Modular Architecture Overview

The system for brain tumor detection is designed with a clean, modular architecture that ensures scalability, flexibility, and ease of maintenance. It consists of four main components:

1. Data Input Module

- **Function**: Handles the upload and preprocessing of MRI scans.
- **Details**: Users upload MRI images, which are first validated and then passed through a preprocessing pipeline. This step ensures data consistency and quality before being fed into the detection engine.

2. Detection Engine

- **Function**: Core inference system powered by the YOLOv11 model.
- **Details**:
    i. Processes preprocessed images to identify and localize brain tumors.
    ii. Uses advanced deep learning layers for real-time, high-precision detection.
    iii. Outputs bounding boxes and confidence scores for each detected tumor region.

3. Visualization Module

- **Function**: Displays detection results interactively using **Streamlit**.
- **Details**:
    i. Highlights detected tumor regions on the MRI scans using bounding boxes.
    ii. Provides real-time visual feedback, enabling users (clinicians or researchers) to examine predictions easily.
    iii. Confidence scores and class labels are overlaid to enhance interpretability.

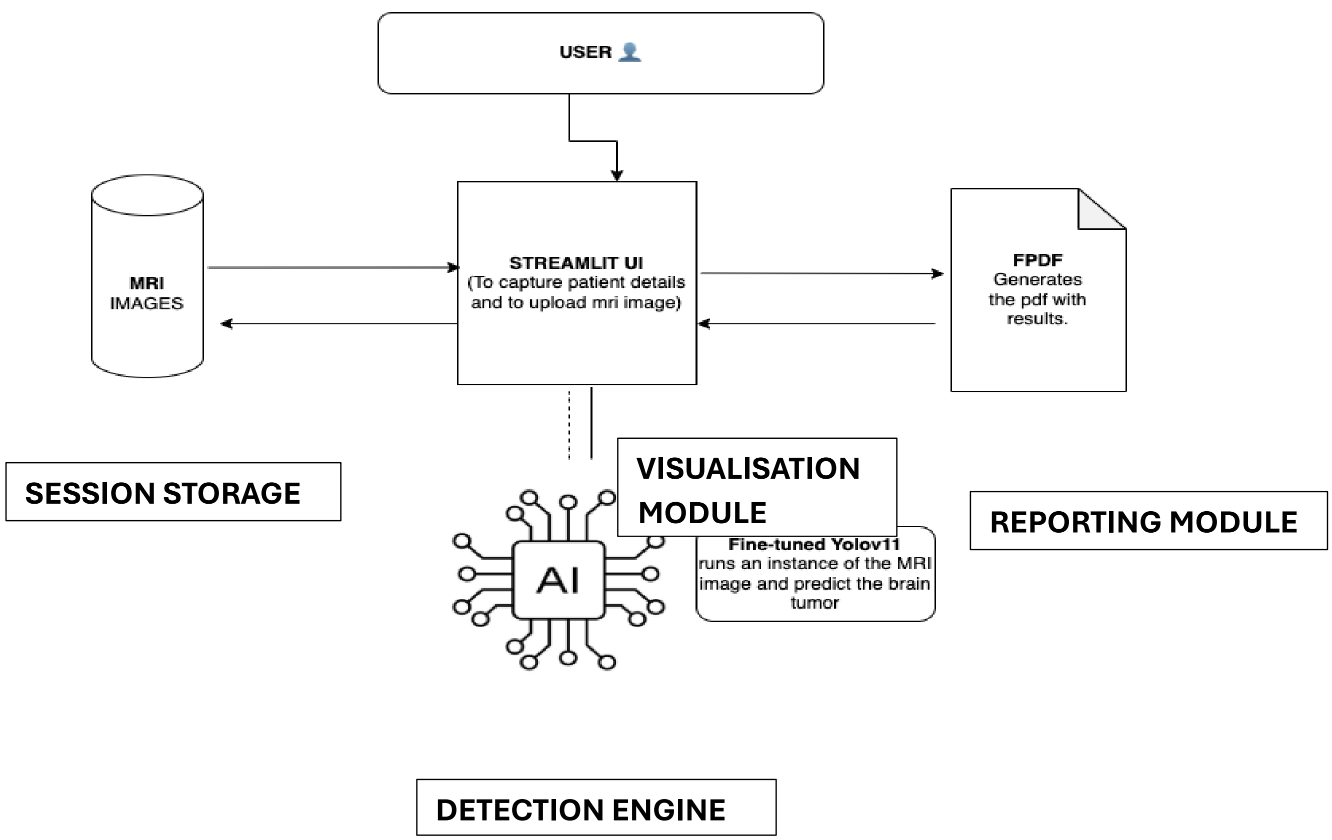
4. Reporting Module

- **Function**: Automatically generates summary reports for detected tumors.
- **Details**:
    i. Utilizes the **FPDF** library to create PDF reports.
    ii. Designed for easy sharing with medical professionals or integration into patient records.

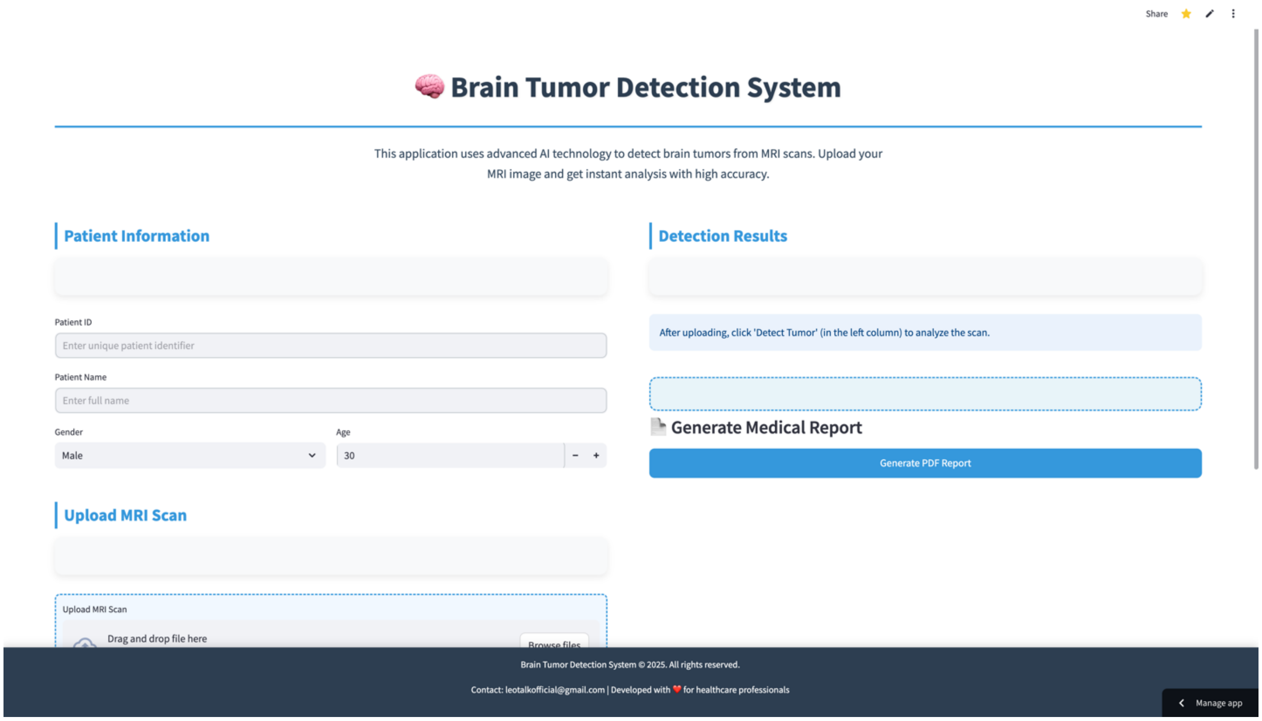
Image Preprocessing Pipeline

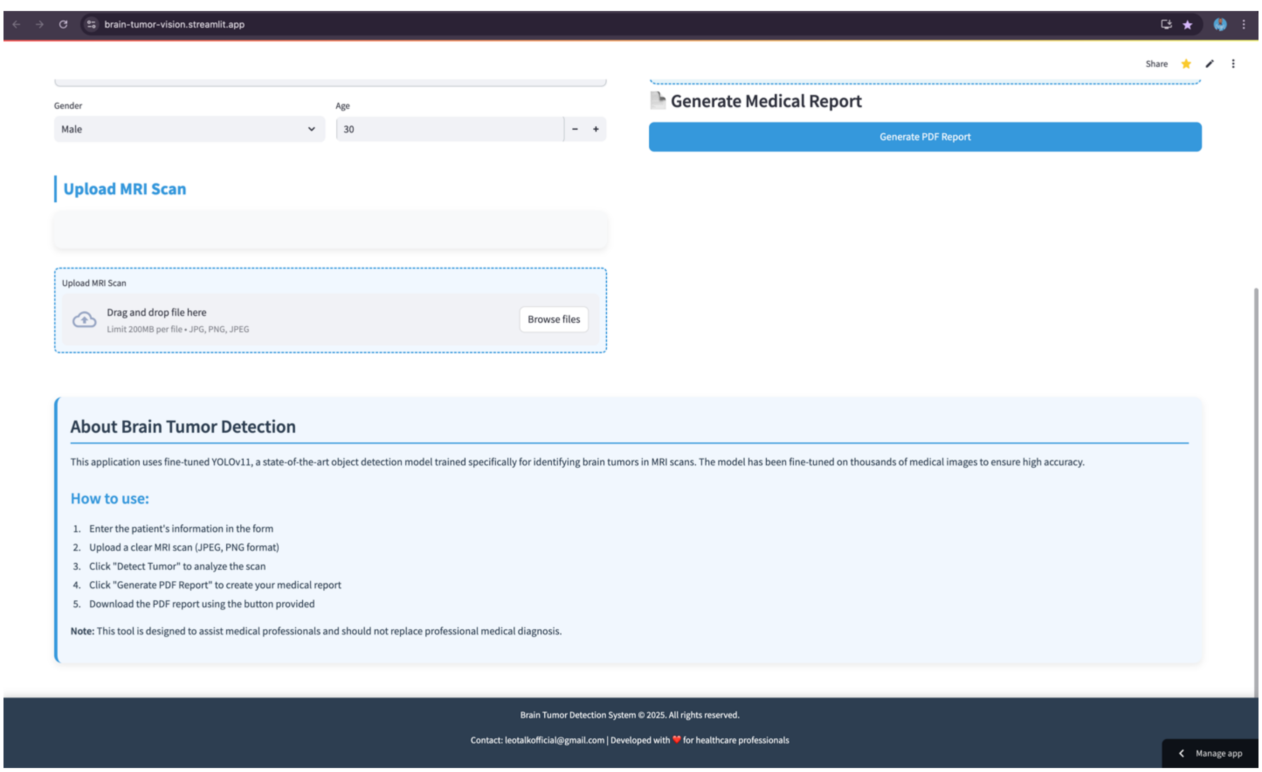
Before inference, uploaded MRI scans go through several transformation steps to standardize input for the model:

- **Resizing**: Adjusts image dimensions to match the input size expected by YOLOv11 (e.g., 640×640).
- **Normalization**: Pixel values are scaled (typically between 0 and 1) to ensure model stability.
- **Tensor Conversion**: Images are converted to tensors and batched appropriately for GPU-based inference.



11. UI Screenshots.





12. User Guide.

1. Enter Patient ID, Patient Name, Gender and Age.
2. Upload an MRI scan image.
3. View analysis with bounding boxes.
4. Generate and download report.

13. Generated Output.

BRAIN TUMOR DETECTION REPORT

Contact: leotalkofficial@gmail.com

Generated on: April 28, 2025 05:14:53

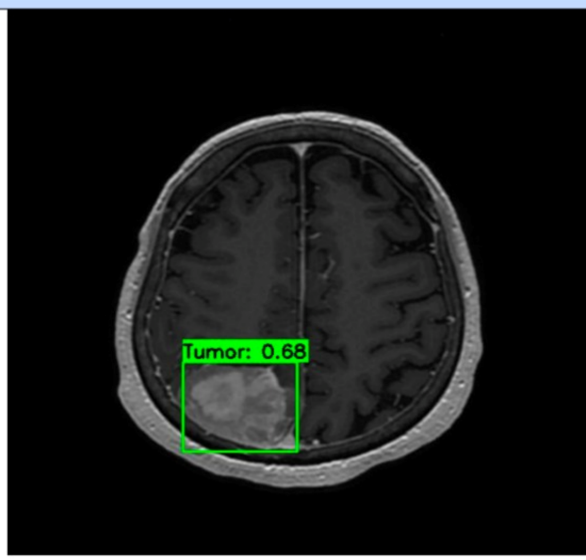
PATIENT INFORMATION	
Patient ID:	ZW-1980-263
Patient Name:	Levi Ragnar
Gender:	Male
Age:	30

DETECTION RESULTS

TUMOR DETECTED

No.	Location	Confidence	Classification
1	(192, 413)	0.68	Benign

MRI SCAN RESULTS



DISCLAIMER: This report is generated using an AI-based detection system and should be reviewed by a licensed medical professional. This tool is designed to aid in diagnosis, not replace professional medical judgment.

14. Conclusion.

This project demonstrates the power of deep learning in the medical domain. With high accuracy and real-time detection, it has the potential to assist radiologists in early tumor diagnosis.

15. Future Enhancements.

1. 3D MRI support.
2. Mobile version.
3. Expert feedback tools.

16. References.

1. Ronneberger et al., U-Net: Biomedical Image Segmentation.
2. https://pjreddie.com/darknet/yolo/.
3. Wang et al., YOLOv11: Real-Time Detection in Medical Imaging.
4. Ultralytics YOLOv11 GitHub - https://github.com/ultralytics/ultralytics.
5. RoboFlow Datasets - https://universe.roboflow.com/.